



Task 1: Environment Setup & Basic Syntax

Description: Set up the Java development environment and write simple programs to grasp core syntax.

Responsibilities:

Install JDK, IDE (IntelliJ/Eclipse), and configure environment variables.

Write 5-10 programs using variables, operators, loops, and conditionals (e.g., Fibonacci series, prime number checker).

Skills Gained:

1. Hello World & Environment Test

Purpose: Validate Java installation and IDE configuration

Concepts Covered:

- Basic class structure (`public class`)
- Main method as entry point (`public static void main`)
- Console output (`System.out.println`)

Learning Objective:

- Verify JDK setup and IDE functionality
 - Understand Java's execution workflow
-

2. Temperature Converter

Purpose: Convert Fahrenheit to Celsius using arithmetic operations

Concepts Covered:

- Variable declaration (`double fahrenheit`)
- Arithmetic operators (`-, *, /`)
- Formatted output (`printf`)

Learning Objective:

- Work with primitive data types
- Understand operator precedence

- Format numerical output
-

3. EvenOdd Checker

Purpose: Determine if a number is even or odd

Concepts Covered:

- Modulo operator (%)
- Ternary operator (condition ? true : false)
- Conditional execution

Learning Objective:

- Make decisions in code
 - Use concise conditional expressions
 - Understand integer division properties
-

4. Multiplication Table

Purpose: Generate multiplication tables using loops

Concepts Covered:

- for loop control structure
- Loop variable initialization/increment
- Formatted string output

Learning Objective:

- Implement repetitive tasks efficiently
 - Control loop iterations
 - Combine variables in output
-

5. Digit Counter

Purpose: Count digits in an integer using while-loop

Concepts Covered:

- while loop with condition
- Integer division ($\div$ 10)
- Counter variable usage

Learning Objective:

- Process numbers digit-by-digit
 - Handle loop termination conditions
 - Modify variables during iteration
-

6. Fibonacci Series

Purpose: Generate number sequence where each number is the sum of the two preceding ones

Concepts Covered:

- Multiple variable assignment
 - Variable value swapping
 - Algorithm implementation
- Learning Objective:
- Solve classic programming problems
 - Manage variable state changes
 - Understand sequence generation logic
-

7. Prime Number Checker

Purpose: Determine if a number is prime

Concepts Covered:

- Boolean flag usage
 - Efficient looping (`Math.sqrt` optimization)
 - break statement for early termination
- Learning Objective:
- Implement mathematical algorithms
 - Optimize loop boundaries
 - Control loop flow with break

8. Day Finder

Purpose: Map integer values to weekday names

Concepts Covered:

- switch statement with arrow syntax
 - Case fall-through prevention
 - Default case handling
- Learning Objective:
- Replace complex if-else chains
 - Use modern switch expressions
 - Handle unexpected inputs
-

9. Array Operations

Purpose: Perform calculations on array elements

Concepts Covered:

- Array initialization (`int[] numbers`)
 - Enhanced for-loop (`for(int n : numbers)`)
 - Aggregate operations (summing)
- Learning Objective:
- Store and process data collections
 - Iterate through arrays
 - Perform batch operations
-

10. Circle Operations

Purpose: Calculate circle properties using methods

Concepts Covered:

- Method declaration (static double)
 - Parameter passing (double r)
 - Code organization with methods
- Learning Objective:
- Modularize code into functions
 - Reuse code through methods
 - Apply mathematical constants (Math.PI)
-

Environment Setup Guide

Step 1: Install JDK

- Download JDK 17+ from [Oracle](#)
- Set JAVA_HOME environment variable
- Verify with terminal: javac -version

Step 2: Install IDE

- Download [IntelliJ IDEA Community Edition](#)
- Configure SDK: File → Project Structure → SDKs → Add JDK

Step 3: First Project

1. Create new Java project
 2. Add Java class: Right-click src → New → Java Class
 3. Type code → Run with green arrow button
-

Skill Development Map

Skill	Practice Method	Success Indicator
Syntax	Type all programs manually (no	0 compilation errors
Mastery	copy-paste)	

Debugging	Intentionally break code and fix errors	Identify 5 common error types
Problem Solving	Modify programs to handle user input	All programs accept dynamic input values
IDE Proficiency	Use keyboard shortcuts for all actions	Complete tasks without mouse
Code Reading	Explain program logic line-by-line	Clear articulation of each code decision

Progression Tips

1. Start Simple: Begin with HelloWorld and TemperatureConverter
2. Add Complexity: Introduce loops (MultiplicationTable) and conditionals (EvenOddChecker)
3. Implement Logic: Solve algorithmic problems (Fibonacci, PrimeChecker)
4. Organize Code: Use methods (CircleOperations) and arrays (ArrayOperations)
5. Experiment:
 - Add Scanner class for user input
 - Implement input validation
 - Convert to OOP style (create Circle class)

Example Enhancement (Prime Checker with Input):

Key Learning Outcomes:

- Transition from static to interactive programs
- Handle real user input
- Practice resource management (`scanner.close()`)

```
// 1. Hello World & Environment Test
```

```
public class HelloWorld {  
  
    public static void main(String[] args) {  
  
        System.out.println("Hello World! JDK setup successful.");  
  
    }  
  
}
```

```
// 2. Variables and Operators
```

```
public class TemperatureConverter {  
  
    public static void main(String[] args) {  
  
        double fahrenheit = 98.6;  
  
        double celsius = (fahrenheit - 32) * 5/9;  
  
        System.out.printf("%.2f°F = %.2f°C\n", fahrenheit, celsius);  
  
    }  
  
}
```

```
// 3. Conditional Statements (If-Else)
```

```
public class EvenOddChecker {  
  
    public static void main(String[] args) {  
  
        int number = 27;  
  
        System.out.println(number % 2 == 0 ? "Even" : "Odd");  
  
    }  
  
}
```

// 4. For Loop

```
public class MultiplicationTable {  
  
    public static void main(String[] args) {  
  
        int num = 5;  
  
        for(int i = 1; i <= 10; i++) {  
  
            System.out.printf("%d x %d = %d\n", num, i, num*i);  
  
        }  
  
    }  
  
}
```

// 5. While Loop

```
public class DigitCounter {  
  
    public static void main(String[] args) {  
  
        int number = 12345, count = 0;  
  
        while(number != 0) {  
  
            number /= 10;  
  
            count++;  
  
        }  
  
        System.out.println("Total digits: " + count);  
  
    }  
  
}
```

// 6. Fibonacci Series

```
public class FibonacciSeries {
```



```

    public static void main(String[] args) {

        int n = 10, t1 = 0, t2 = 1;

        System.out.print("Fibonacci Series: ");

        for (int i = 1; i <= n; i++) {

            System.out.print(t1 + " ");

            int sum = t1 + t2;

            t1 = t2;

            t2 = sum;

        }

    }
}

```

// 7. Prime Number Checker

```

public class PrimeChecker {

    public static void main(String[] args) {

        int num = 29;

        boolean isPrime = true;

        for(int i = 2; i <= Math.sqrt(num); i++) {

            if(num % i == 0) {

                isPrime = false;

                break;

            }

        }

        System.out.println(num + (isPrime ? " is prime" : " is not prime"));
    }
}

```

```
    }  
}
```

```
// 8. Switch Statement
```

```
public class DayFinder {  
    public static void main(String[] args) {  
        int day = 3;  
        switch(day) {  
            case 1 -> System.out.println("Monday");  
            case 2 -> System.out.println("Tuesday");  
            case 3 -> System.out.println("Wednesday");  
            default -> System.out.println("Invalid day");  
        }  
    }  
}
```

```
// 9. Array Processing
```

```
public class ArrayOperations {  
    public static void main(String[] args) {  
        int[] numbers = {5, 2, 9, 1, 5};  
        int sum = 0;  
        for(int n : numbers) {  
            sum += n;  
        }  
    }  
}
```

```
        System.out.println("Array sum: " + sum);  
    }  
}
```

// 10. Method Usage

```
public class CircleOperations {  
  
    public static void main(String[] args) {  
  
        double radius = 7.5;  
  
        System.out.printf("Circumference: %.2f\n",  
calculateCircumference(radius));  
  
        System.out.printf("Area: %.2f\n", calculateArea(radius));  
  
    }  
  
    static double calculateCircumference(double r) {  
  
        return 2 * Math.PI * r;  
  
    }  
  
    static double calculateArea(double r) {  
  
        return Math.PI * r * r;  
  
    }  
}
```